

Deployment Status - Bookstore Microservices Application

All Containers Running Successfully

Container Status (10 Services Total)

CONTAINER ID	IMAGE	STATUS	PORTS
76542a111add	bookstore-app-order-service	Up 8 minutes	0.0.0.0:5006->5000/tcp
3708fbf6c442	bookstore-app-book-service	Up 8 minutes	0.0.0.0:5003->5000/tcp
956223b31117	bookstore-app-user-service	Up 8 minutes	0.0.0.0:5002->5000/tcp
94d09594de80	bookstore-app-auth-service	Up 8 minutes	0.0.0.0:5001->5000/tcp
6f924400c25b	bookstore-app-review-service	Up 8 minutes	0.0.0.0:5004->5000/tcp
f599c69a9ded	bookstore-app-frontend-service	Up 8 minutes	0.0.0.0:8080->5000/tcp
3fc0f35c3259	mysql:5.7	Up 8 minutes	0.0.0.0:3306->3306/tcp
bc2e2034df1f	bookstore-app-cart-service	Up 8 minutes	0.0.0.0:5005->5000/tcp
f570a6aa5e93	bookstore-app-payment-service	Up 8 minutes	0.0.0.0:5007->5000/tcp
f0fd32ae8474	bookstore-app-notification-service	Up 8 minutes	0.0.0.0:5008->5000/tcp

Access URLs (All Verified Working)

Frontend Dashboard

- **Main Frontend:** <http://localhost:8080>
 - Displays links to all microservices
 - HTML interface with clickable links

Microservices (JSON APIs)

1. **Auth Service:** <http://localhost:5001>
 - Response: {"service": "auth-service", "message": "auth-service is running"}
2. **User Service:** <http://localhost:5002>
 - Response: {"service": "user-service", "message": "user-service is running"}
3. **Book Service:** <http://localhost:5003>
 - Response: {"service": "book-service", "message": "book-service is running"}
4. **Review Service:** <http://localhost:5004>
 - Response: {"service": "review-service", "message": "review-service is running"}
5. **Cart Service:** <http://localhost:5005>
 - Response: {"service": "cart-service", "message": "cart-service is running"}
6. **Order Service:** <http://localhost:5006>
 - Response: {"service": "order-service", "message": "order-service is running"}

7. **Payment Service:** http://localhost:5007
 - Response: {"service": "payment-service", "message": "payment-service is running"}
8. **Notification Service:** http://localhost:5008
 - Response: {"service": "notification-service", "message": "notification-service is running"}

Database

9. **MySQL Database:** localhost:3306
 - Database: bookstore
 - Root Password: root
 - Persistent volume: db_data

Commands Used

Start All Services

```
cd /home/maaz/uni_work_semester7/Assignment_1_DevOps/bookstore-app
sudo docker compose up --build
```

Stop All Services

```
sudo docker compose down
```

View Running Containers

```
sudo docker ps
```

View Container Logs

```
sudo docker compose logs -f [service-name]
```

Testing Instructions

Test Frontend

```
curl http://localhost:8080
# Or open in browser: http://localhost:8080
```

Test Individual Services

```
curl http://localhost:5001 # Auth
curl http://localhost:5002 # User
curl http://localhost:5003 # Book
curl http://localhost:5004 # Review
curl http://localhost:5005 # Cart
curl http://localhost:5006 # Order
```

```
curl http://localhost:5007 # Payment
curl http://localhost:5008 # Notification
```

Screenshots for Assignment Report

1. **Docker PS Output:** Shows all 10 containers running
2. **Frontend Browser View:** <http://localhost:8080> showing service links
3. **Individual Service Responses:** JSON responses from each microservice
4. **Docker Compose Build:** Terminal output showing successful build

Architecture Overview

Frontend Service
(Port 8080)

Auth Service
(Port 5001)

User Service
(Port 5002)

Book Service
(Port 5003)

MySQL DB
(Port 3306)

Additional Services:

- Review Service (5004)
- Cart Service (5005)
- Order Service (5006)
- Payment Service (5007)
- Notification Service (5008)

Project Structure

```
bookstore-app/
  docker-compose.yml    # Orchestration configuration
  README.md             # Project documentation
  auth-service/         # Authentication microservice
  user-service/         # User management
  book-service/         # Book catalog
  review-service/       # Book reviews
```

cart-service/	# Shopping cart
order-service/	# Order processing
payment-service/	# Payment handling
notification-service/	# Notifications
frontend-service/	# Web interface

Success Criteria

- ☒ 10 services created (9 Flask + 1 MySQL)
- ☒ Each service in separate folder
- ☒ Each service has app.py, requirements.txt, Dockerfile
- ☒ docker-compose.yml configured
- ☒ All containers build successfully
- ☒ All containers running simultaneously
- ☒ Frontend accessible at <http://localhost:8080>
- ☒ All microservices respond with JSON
- ☒ Services can be accessed via browser/curl

Next Steps for Assignment Report

1. Take screenshot of `docker ps` output
2. Take screenshot of frontend in browser (<http://localhost:8080>)
3. Take screenshots of 2-3 microservice JSON responses
4. Create architecture diagram (provided above as template)
5. Document the purpose of each microservice
6. Explain Docker Compose configuration
7. Include commands and testing procedures

Generated: 2025-10-13T23:29:34+05:00 **Status:** All services operational